

mb-pc40s-2020-jan

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

Determine and simplify the 8th term in the binomial expansion of $\left(x - \frac{2}{x^3}\right)^{10}$.

Solution

$$\begin{aligned}t_8 &= {}_{10}C_7(x)^3\left(-\frac{2}{x^3}\right)^7 \quad 2 \text{ marks (1 mark for } {}_{10}C_7; \frac{1}{2} \text{ mark for each consistent factor)} \\&= 120(x^3)\left(-\frac{128}{x^{21}}\right) \\&= -\frac{15360}{x^{18}} \quad 1 \text{ mark for simplification } \left(\frac{1}{2} \text{ mark for coefficient; } 1/2 \text{ mark for exponent}\right)\end{aligned}$$

3 marks

Question 2

The temperature of hot chocolate is recorded as it cools down. The data shows that the temperature cools according to the equation:

$$T = 82(0.87)^t + 16$$

where T is the temperature of the hot chocolate, in degrees Celsius after t minutes and t is the time, in minutes, after the hot chocolate is made.

In order to avoid burns, hot chocolate should not be served at a temperature higher than 71°C .

Determine, algebraically, the amount of time it will take the hot chocolate to reach this temperature.

Solution

$$71 = 82(0.87)^t + 16$$

$$55 = 82(0.87)^t$$

$$\frac{55}{82} = 0.87^t$$

$$\log\left(\frac{55}{82}\right) = t \log(0.87)$$

$\frac{1}{2}$ mark for applying logarithms

1 mark for power law

$$t = \frac{\log\left(\frac{55}{82}\right)}{\log(0.87)}$$

$$t = 2.867874$$

$$t = 2.868 \text{ minutes}$$

$\frac{1}{2}$ mark for evaluating quotient of logarithms

2 marks

Question 3

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$3 \sin^2 \theta + 6 \sin \theta + 2 = 0$$

Solution

$$\sin \theta = \frac{-6 \pm \sqrt{6^2 - 4(3)(2)}}{2(3)}$$

$$\sin \theta = \frac{-6 \pm \sqrt{12}}{6}$$

$$\sin \theta = -0.422\ 649\dots$$

$$\theta_r = 0.436367\dots$$

$$\theta = 3.578, 5.847$$

$\frac{1}{2}$ mark for substitution into quadratic formula

$\frac{1}{2}$ mark for solving for $\sin \theta$

2 marks for solving for θ

($\frac{1}{2}$ mark for each value; 1 mark for indicating
no solution)

3 marks

Question 4

A committee of 4 people is to be selected from 6 high school students and 5 middle years students.

Determine the number of possible committees if the committee must have at least 3 high school students.

Solution

Case 1:

$${}_6C_3 \cdot {}_5C_1 = 100$$

Case 2:

$${}_6C_4 \cdot {}_5C_0 = 15$$

$100 + 15 = 115$ possible committees

1 mark for Case 1

1 mark for Case 2

1 mark for addition of cases

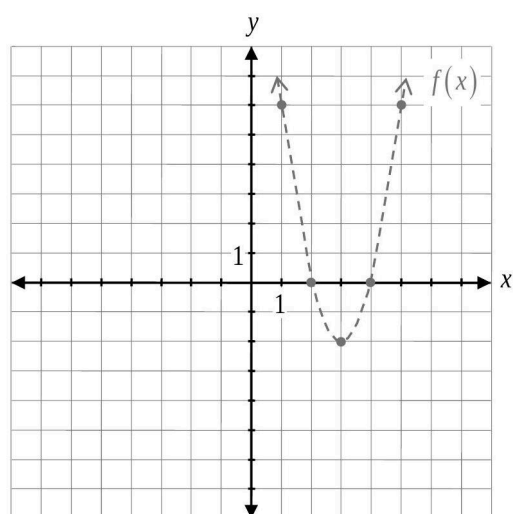
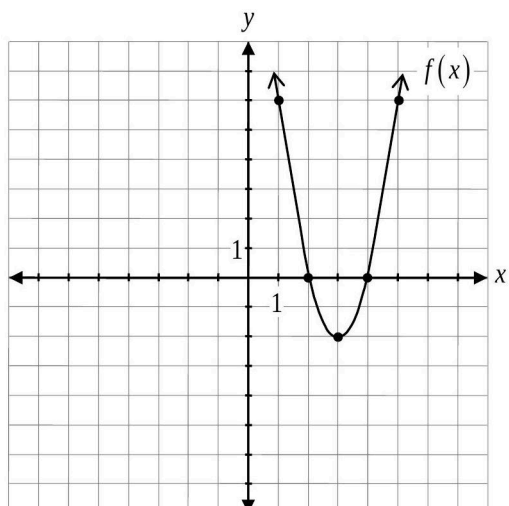
3 marks

Note:

- ${}_5C_0$ does not need to be shown.

Question 5

Given the graph of $y = f(x)$, sketch the graph of $y = -|f(x)|$.

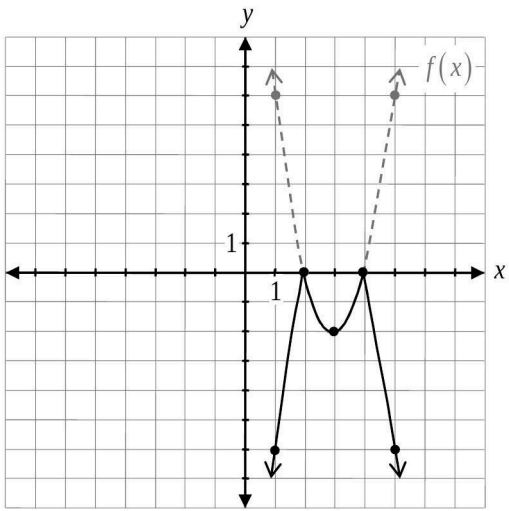


The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Solution

Solution



1 mark for absolute value

1 mark for vertical reflection

2 marks

Question 6

Describe the transformations used to obtain the graph of the function $y = 4f\left(\frac{x}{2}\right) - 3$ from the graph of $y = f(x)$.

Solution

The graph of $y = f(x)$ is vertically stretched by a factor of 4, horizontally stretched by a factor of 2, and translated down 3 units.

1 mark for vertical stretch

1 mark for horizontal stretch

1 mark for vertical translation

3 marks

Note:

- Deduct a maximum of 1 mark for concept error of incorrect order of vertical transformations.

Question 7

If $\log 4 = m$ and $\log 3 = n$, express $\log 48$ in terms of m and n .

Solution

$$\begin{aligned}\log 48 &= \log(4^2 \cdot 3) \\ &= 2 \log 4 + \log 3 \\ &= 2m + n\end{aligned}$$

1 mark for power law

1 mark for product law

2 marks

Question 8

State the domain and range of the radical function, $f(x) = -3\sqrt{x-8} + 1$.

Domain:

Range:

Solution

1 mark for domain

Range: $(-\infty, 1]$

1 mark for range

2 marks

Question 9

Explain why the graph of $y = \tan x$ does not have an amplitude.

Solution

The range of $y = \tan x$ is all real numbers.

1 mark

or

As the graph of $y = \tan x$ approaches the vertical asymptotes, the function value approaches positive and negative infinity.

Question 10

Prove the following identity for all permissible values of θ .

$$\frac{\sec \theta - \sin^2 \theta \sec \theta}{\tan \theta \sin \theta} = \csc^2 \theta - 1$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{1}{\left(\frac{\sin \theta}{\cos \theta}\right)(\sin \theta)} - \frac{\sin^2 \theta}{\cos \theta}$ $\frac{\frac{1 - \sin^2 \theta}{\cos \theta}}{\frac{\sin^2 \theta}{\cos \theta}}$ $\frac{\cos^2 \theta}{\sin^2 \theta}$ $\cot^2 \theta$ $\csc^2 \theta - 1$	$\csc^2 \theta - 1$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 11

Given that $x = 3$ is one of the zeros of $p(x) = x^3 - 7x - 6$, express $p(x)$ in completely factored form.

$p(x) =$

Solution

3	1	0	-7	-6
		3	9	6
	1	3	2	0

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x - 3)(x^2 + 3x + 2)$$
$$p(x) = (x - 3)(x + 2)(x + 1)$$

1/2 mark for factor of $(x - 3)$
1/2 mark for other consistent factors

2 marks

Question 12

Solve, algebraically.

$${}_{n+1}P_2 = 6$$

Solution

Using the permutation formula,

$$\begin{aligned}\frac{(n+1)!}{(n-1)!} &= 6 \\ \frac{(n+1)n(n-1)!}{(n-1)!} &= 6 \\ n(n+1) &= 6 \\ n^2 + n - 6 &= 0 \\ (n+3)(n-2) &= 0\end{aligned}$$

The algebraic roots are $n = -3$ and $n = 2$. Reject $n = -3$: the number of objects must allow a selection of two, so $n + 1 \geq 2$.

Therefore $n = 2$. Check: $3 \times 2 = 6$.

Scoring: $\frac{1}{2}$ mark for substitution; 1 mark for factorial expansion; $\frac{1}{2}$ mark for simplification; $\frac{1}{2}$ mark for the permissible value; $\frac{1}{2}$ mark for rejecting the extraneous root.

Question 13

Express $\log(3x - 1) - \log x + \log 9$ as a single logarithm.

Solution

$$\log\left(\frac{9(3x-1)}{x}\right)$$

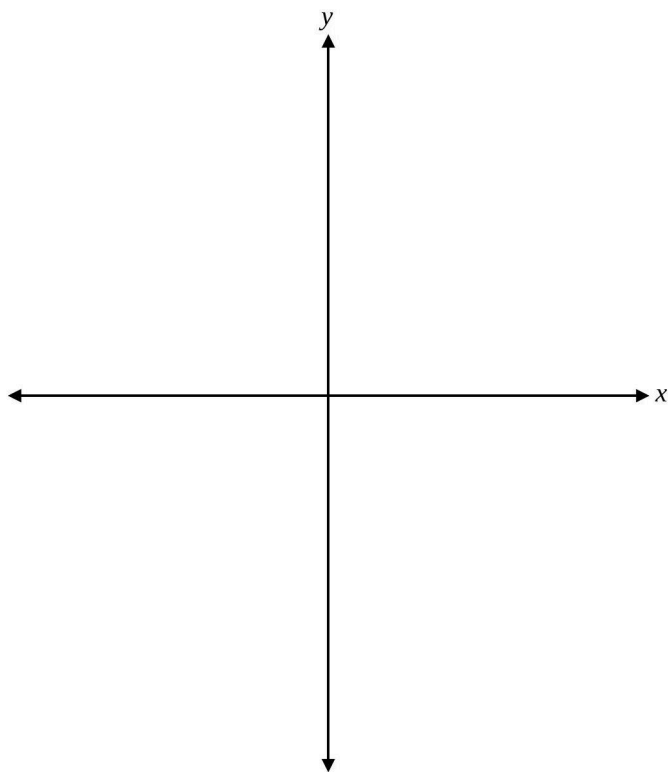
1 mark for product law

1 mark for quotient law

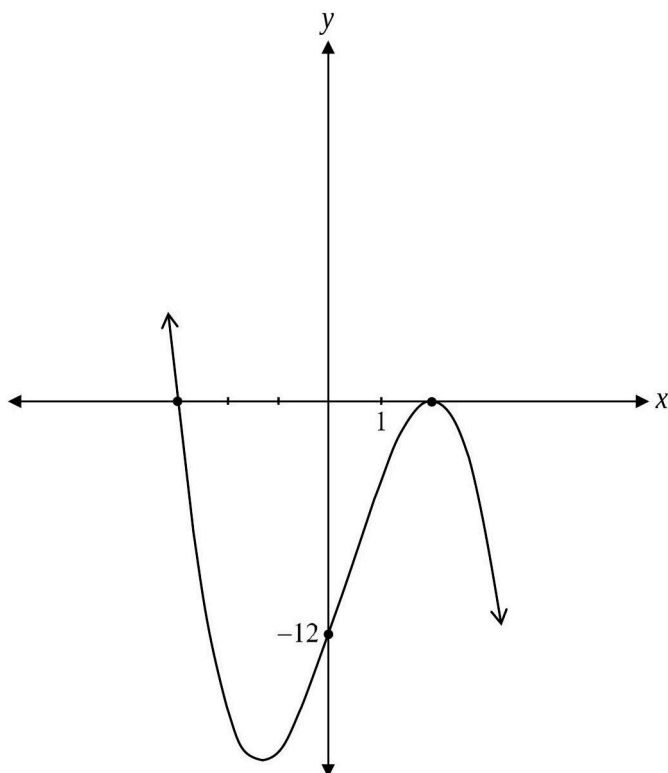
2 marks

Question 14

Sketch the graph of $P(x) = -(x - 2)^2(x + 3)$.



Solution



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity of 2 at $x = 2$

$\frac{1}{2}$ mark for end behaviour

3 marks

Question 15

Given $f(x) = -2x - 5$ and $g(x) = x + 7$, identify the equation of $h(x) = f(g(x))$.

A.

$$h(x) = -2x^2 - 35$$

B.

$$h(x) = -2x - 19$$

C.

$$h(x) = -2x^2 - 19x - 35$$

D.

$$h(x) = -2x + 2$$

Solution

Correct answer: B

$$h(x) = -2x - 19$$

Question 16

The function $y = f(x)$ has a domain of $[2,5]$ and a range of $[-6,3]$. Identify the range of the function $y = f^{-1}(x)$.

A.

$[-6, 3]$

B.

$[-5, 2]$

C.

$[-3, 6]$

D.

$[2, 5]$

Solution

Correct answer: D

$[2, 5]$

Question 17

Identify the function that has a graph with a point of discontinuity (hole) at $x = -3$.

A.

$$y = \frac{x+3}{x^2-9}$$

B.

$$y = \frac{x-3}{x^2-9}$$

C.

$$y = \frac{x^2-9}{x-3}$$

D.

$$y = \frac{x^2+9}{x+3}$$

Solution

Correct answer: A

$$y = \frac{x+3}{x^2-9}$$

Question 18

Identify the value of $\ln e$.

A.

0

B.

$\log e$

C.

1

D.

e

Solution

Correct answer: C

1

Question 19

When the polynomial function, $p(x)$, is divided by $(x - 4)$ the remainder is 17 . Identify which statement is true.

A.

$$p(4) = 17$$

B.

$$p(-4) = 17$$

C.

$$p(4) = 0$$

D.

$$p(-4) = 0$$

Solution

Correct answer: A

$$p(4) = 17$$

Question 20

Identify the expression equivalent to $\frac{(n-6)!}{(n-4)!}$.

A.

$$(n-4)(n-5)$$

B.

$$\frac{1}{(n-4)(n-5)}$$

C.

$$(n-6)(n-5)$$

D.

$$\frac{1}{(n-6)(n-5)}$$

Solution

Correct answer: B

$$\frac{1}{(n-4)(n-5)}$$

Question 21

Identify the quadrant in which θ terminates if $\sec \theta = -\frac{4}{3}$ and $\sin \theta > 0$.

A.

I

B.

II

C.

III

D.

IV

Solution

Correct answer: B

II

Question 22

Identify the expression that represents all angles that are coterminal with $\frac{\pi}{3}$.

A.

$$\frac{\pi}{3} + \pi k, k \in \mathbb{Z}$$

B.

$$\frac{\pi}{3} + \pi k, k \in \mathbb{R}$$

C.

$$\frac{\pi}{3} + 2\pi k, k \in \mathbb{Z}$$

D.

$$\frac{\pi}{3} + 2\pi k, k \in \mathbb{R}$$

Solution

Correct answer: C

$$\frac{\pi}{3} + 2\pi k, k \in \mathbb{Z}$$

Question 23

Given $f(x) = \frac{1}{2}x - 3$, state the coordinates of an invariant (unchanged) point when sketching the graph of $y = \sqrt{f(x)}$.

Solution

$$\begin{aligned}\frac{1}{2}x - 3 &= 0 \\ \frac{1}{2}x &= 3 \\ x &= 6 \\ (6, 0)\end{aligned}$$

1 mark

or

$$\begin{aligned}\frac{1}{2}x - 3 &= 1 \\ \frac{1}{2}x &= 4 \\ x &= 8 \\ (8, 1)\end{aligned}$$

Question 24

Solve, algebraically.

$$125^{3x+4} = \left(\frac{1}{5}\right)^x$$

Solution

$$(5^3)^{3x+4} = (5^{-1})^x$$

$$5^{9x+12} = 5^{-x}$$

$$9x + 12 = -x$$

$$10x = -12$$

$$x = -\frac{12}{10}$$

or

$$x = -\frac{6}{5}$$

1 mark for changing to a common base ($\frac{1}{2}$ mark for each side)

$\frac{1}{2}$ mark for exponent law

$\frac{1}{2}$ mark for equating exponents

2 marks

Question 25

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ and $\cos \theta = -\frac{\sqrt{2}}{2}$, justify that the value of $\tan(2\theta)$ is undefined.

Solution

Method 1

$$\begin{aligned}\tan(2\theta) &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \\ &= \frac{2(1)}{1 - 1} \\ &= \frac{2}{0} \\ &\therefore \text{undefined}\end{aligned}$$

1 mark for value of $\tan \theta$

1 mark for justification

2 marks

Method 2

$$\begin{aligned}\tan \theta &= 1 \\ \theta &= \frac{5\pi}{4} \\ \tan(2\theta) &= \tan\left(2\left(\frac{5\pi}{4}\right)\right) \\ &= \tan \frac{5\pi}{2} \\ &\therefore \text{undefined}\end{aligned}$$

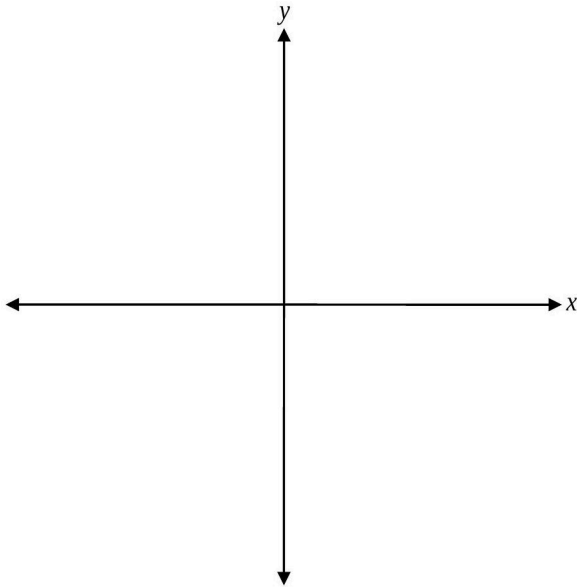
1 mark for value of θ

1 mark for justification

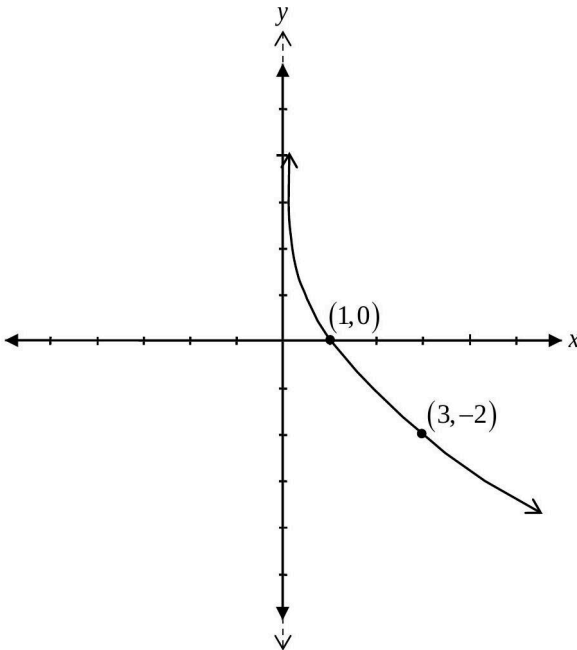
2 marks

Question 26

Sketch the graph of $y = -2 \log_3 x$.



Solution



1 mark for asymptotic behaviour

approaching $x = 0$

1 mark for vertical reflection

1 mark for vertical stretch

3 marks

Question 27

Evaluate.

$$\cos\left(\pi \cdot \sin\left(-\frac{\pi}{6}\right)\right)$$

Solution

$\cos\left(\pi\left(-\frac{1}{2}\right)\right)$ 1 mark for $\sin\left(-\frac{\pi}{6}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

$\cos\left(-\frac{\pi}{2}\right)$

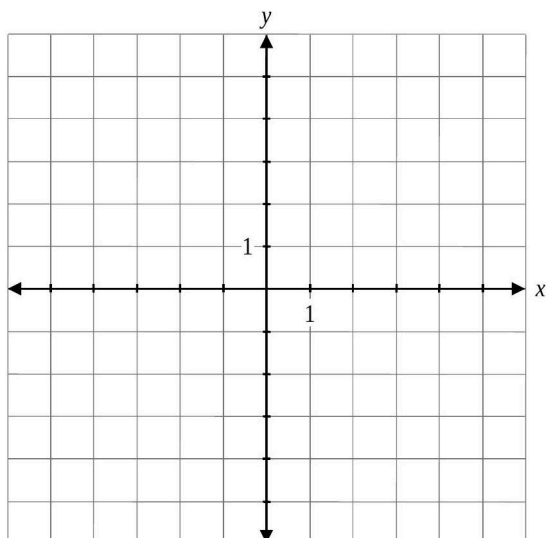
1 mark for consistent value

0

2 marks

Question 28

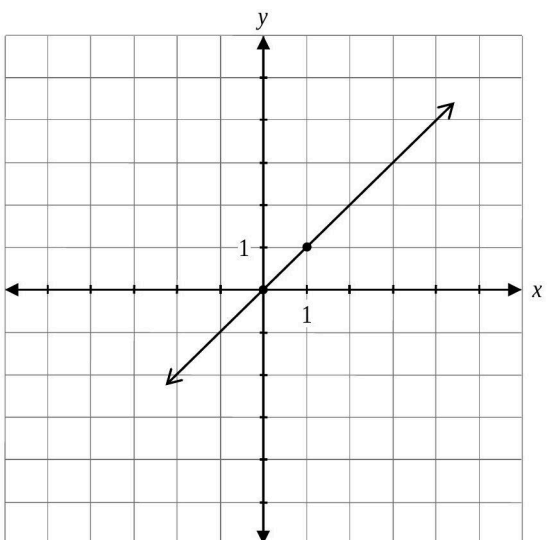
a) Given $f(x) = -x + 2$, sketch the graph of $h(x) = f(f(x))$.



b) Explain why the domain of $h(x) = f(f(x))$ does not have any restrictions.

Solution

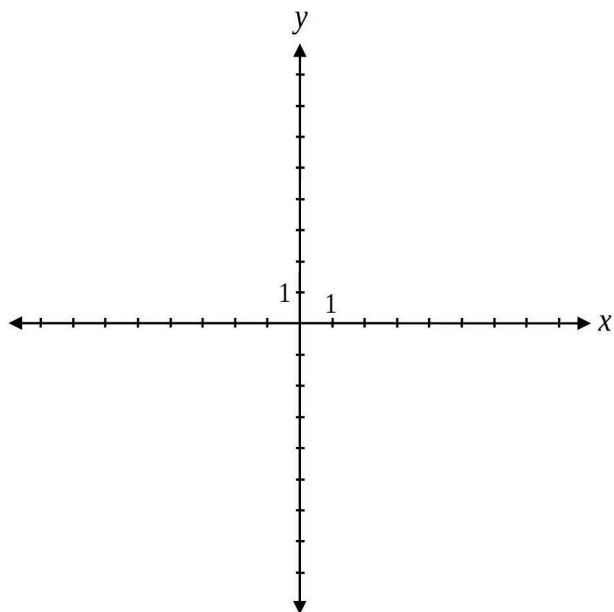
a)



b) Since the domain of $y = f(x)$ does not have any restrictions, then the domain of the composition does not have any restrictions.

Question 29

a) Sketch the graph of $f(x) = \frac{3x-5}{2x+4}$.

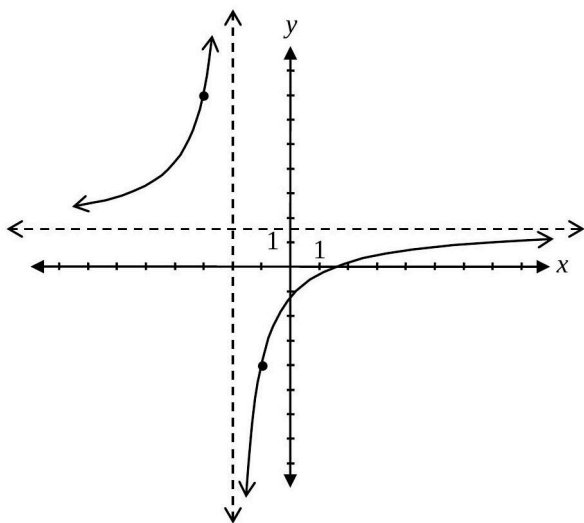


b) State the range of $f(x)$.

Range:

—

Solution



1 mark for asymptotic behaviour approaching $x = -2$

1 mark for asymptotic behaviour approaching $y = \frac{3}{2}$

$\frac{1}{2}$ mark for graph right of $x = -2$

$\frac{1}{2}$ mark for graph left of $x = -2$

3 marks

b) Range: $\{y \in \mathbb{R} \mid y \neq \frac{3}{2}\}$

1 mark for range consistent with graph

1 mark

Question 30

Determine, algebraically, the equation of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 4
- $p(x)$ has a zero at 3 with a multiplicity of 2
- $p(x)$ has zeroes at -1 and -2
- $p(x)$ passes through the point $(2, 24)$.

$p(x) =$

Solution

	$\frac{1}{2}$ mark for factors of $p(x)$
$p(x) = a(x + 1)(x + 2)(x - 3)^2$	$\frac{1}{2}$ mark for multiplicity of 2 at $x = 3$
$24 = a(2 + 1)(2 + 2)(2 - 3)^2$ $24 = 12a$	$\frac{1}{2}$ mark for substitution of $p(2) = 24$
$a = 2$	$\frac{1}{2}$ mark for consistent value of a
$p(x) = 2(x + 1)(x + 2)(x - 3)^2$	2 marks

Question 31

Verify, by substitution, that the equation $\frac{\cos \theta + \cot \theta}{\cot \theta} = 1 + \sin \theta$ is true for $\theta = \frac{2\pi}{3}$.

Left-Hand Side	Right-Hand Side

Solution

Left-Hand Side	Right-Hand Side
$\frac{\cos\left(\frac{2\pi}{3}\right) + \cot\left(\frac{2\pi}{3}\right)}{\cot\left(\frac{2\pi}{3}\right)}$ $\frac{-\frac{1}{2} - \frac{1}{\sqrt{3}}}{-\frac{1}{\sqrt{3}}}$ $\frac{\frac{-\sqrt{3}-2}{2\sqrt{3}}}{-\frac{1}{\sqrt{3}}}$ $\frac{-\sqrt{3}-2}{-2}$ $\frac{-(\sqrt{3}+2)}{-2}$ $\frac{\sqrt{3}+2}{2}$	$1 + \sin\left(\frac{2\pi}{3}\right)$ 1/2 mark for substitution $1 + \frac{\sqrt{3}}{2}$ 1 1/2 marks for exact values (1/2 mark for each) $\frac{2}{2} + \frac{\sqrt{3}}{2}$ $\frac{2 + \sqrt{3}}{2}$ $\frac{\sqrt{3} + 2}{2}$ 1 mark for simplification (1/2 mark for LHS; 1/2 mark for RHS) 3 marks

Note:

- Deduct a maximum of 1 mark if student proves identity without substitution of $\theta = \frac{2\pi}{3}$.

Question 32

Jyugo was asked to state the equation of the vertical asymptote(s) on the graph of $f(x) = \frac{3x-15}{x^2-5x}$.

His solution:

$$\begin{array}{l} f(x) = \frac{3(x-5)}{x(x-5)} \\ x = 0 \quad x = 5 \end{array}$$

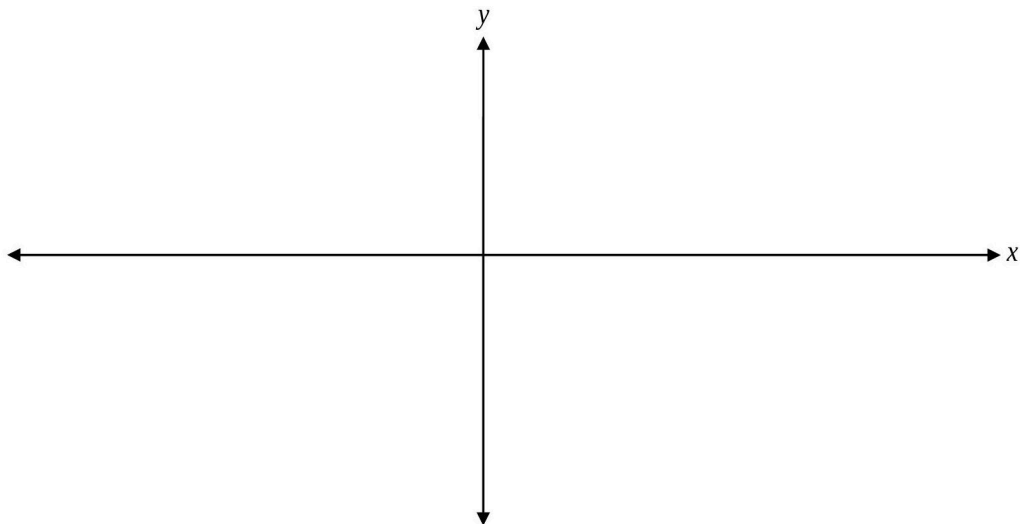
Explain why his solution is incorrect.

Solution

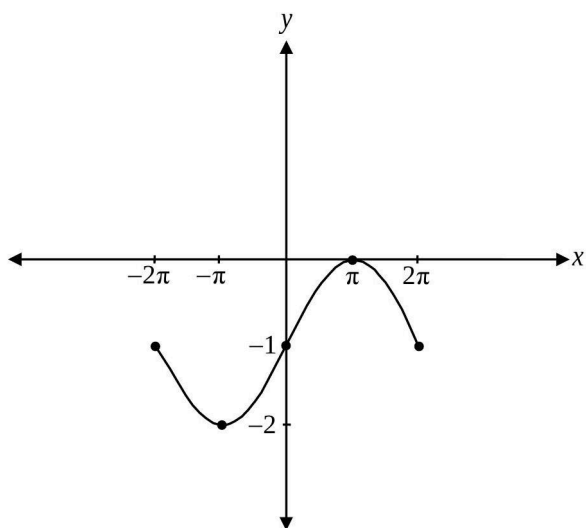
His solution at $x = 5$ is the x -value of a point of discontinuity (hole), not a vertical asymptote.

Question 33

Sketch the graph of $y = \sin\left(\frac{1}{2}x\right) - 1$ over the interval $[-2\pi, 2\pi]$.



Solution



1 mark for shape of $y = \sin x$

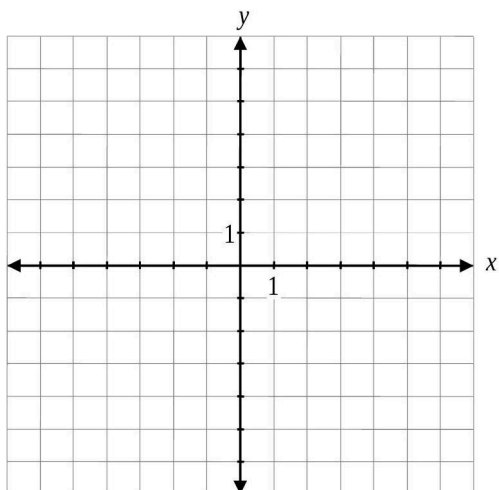
1 mark for period

1 mark for vertical translation

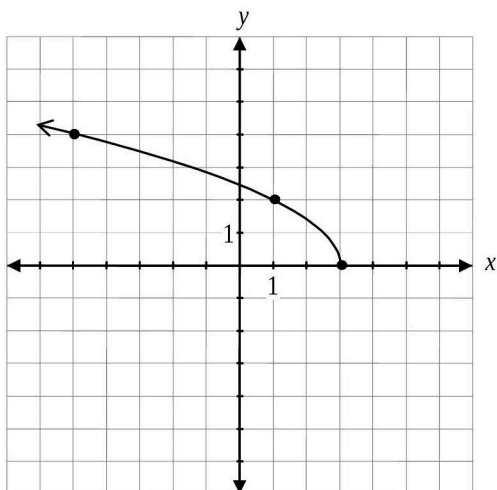
3 marks

Question 34

Sketch the graph of $y = \sqrt{-2x + 6}$.



Solution



1 mark for shape of a radical function

1 mark for horizontal reflection

1 mark for horizontal compression

1 mark for horizontal translation

4 marks

Question 35

Determine the measure of angle R , in radians.

Solution

$$(110)\left(\frac{\pi}{180}\right) = \frac{11\pi}{18} \quad 1 \text{ mark for conversion}$$

$$\angle R = \pi - \left(\frac{\pi}{9} + \frac{11\pi}{18}\right)$$

$$\angle R = \frac{18\pi}{18} - \left(\frac{2\pi}{18} + \frac{11\pi}{18}\right)$$

$$\angle R = \frac{5\pi}{18}$$

1 mark for angle R

2 marks

Question 36

Given $f(x) = \sqrt{x-4}$, state the equation of the resulting function, $g(x)$, after a reflection over the x -axis.

$$g(x) =$$

Solution

$$g(x) = -\sqrt{x-4} \quad \text{1 mark for vertical reflection}$$

1 mark

Question 37

a) Determine the coterminal angle of $\frac{29\pi}{12}$ over the interval $[0, 2\pi]$.

b) Determine the exact value of $\sin\left(\frac{29\pi}{12}\right)$.

Solution

a) $\frac{5\pi}{12}$

1 mark

b)

$$\begin{aligned}\sin\left(\frac{5\pi}{12}\right) &= \sin\left(\frac{\pi}{6} + \frac{\pi}{4}\right) \\ &= \sin\frac{\pi}{6} \cos\frac{\pi}{4} + \cos\frac{\pi}{6} \sin\frac{\pi}{4} \\ &= \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) \\ &= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4} \\ &= \frac{\sqrt{2} + \sqrt{6}}{4}\end{aligned}$$

or

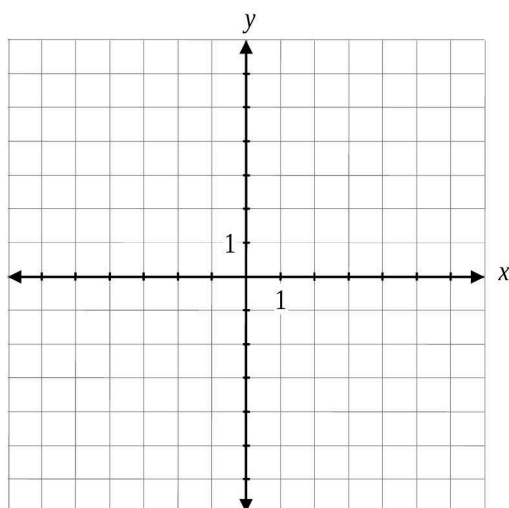
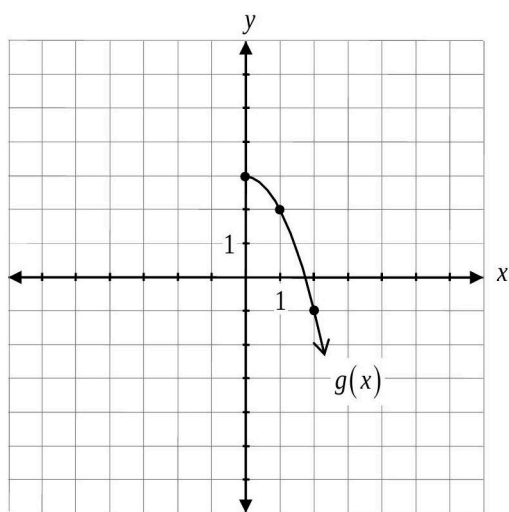
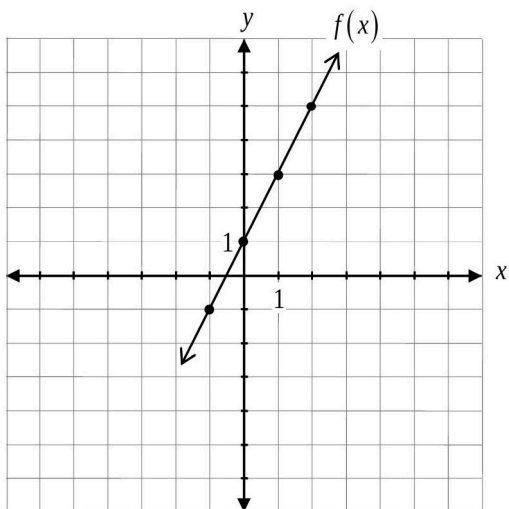
$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

Note:

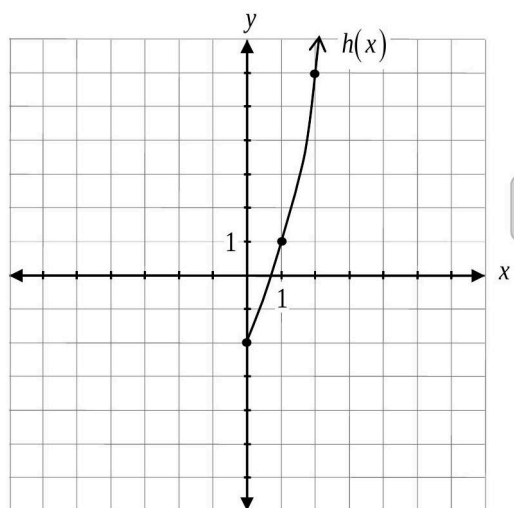
- Other combinations are possible.

Question 38

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) - g(x)$.



Solution



1 mark for operation of subtraction

1 mark for restricted domain

2 marks

Question 39

Determine an equation of a sinusoidal function that has the following characteristics:

- an amplitude of 3
- a period of 6
- a minimum value of -5

Solution

$$y = A \sin(Bx) + D$$

$$y = 3 \sin\left(\frac{\pi}{3}x\right) - 2$$

or

$$y = A \cos(Bx) + D$$

$$y = 3 \cos\left(\frac{\pi}{3}x\right) - 2$$

1 mark for value of A

1 mark for value of B

1 mark for value of D

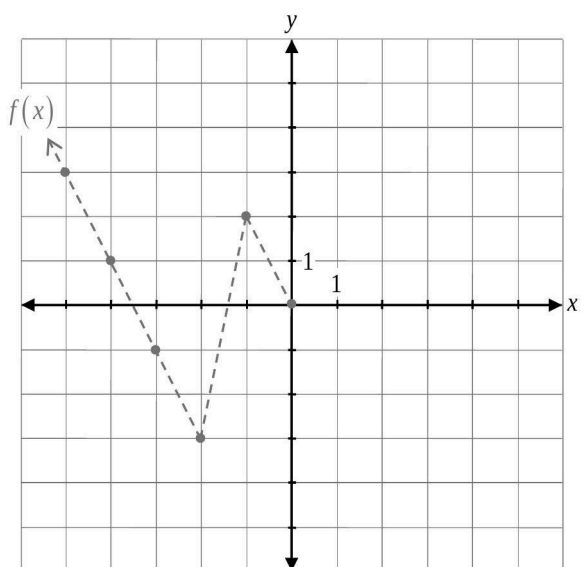
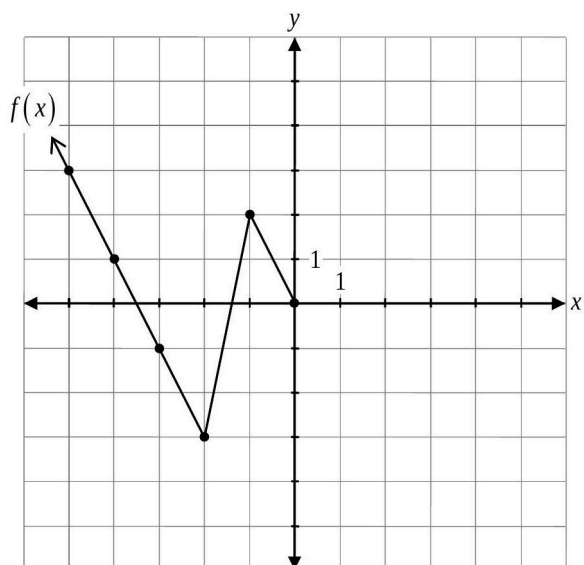
3 marks

Note:

- Other equations are possible.

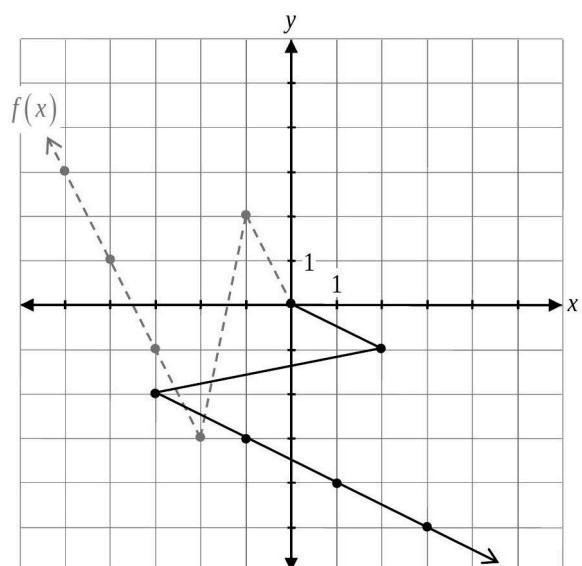
Question 40

Given the graph of $y = f(x)$, sketch the graph after a reflection over the line $y = x$.



The graph of
 $f(x)$ has already
been drawn for
your reference.
No marks will be
awarded for the
graph of $f(x)$.

Solution



1 mark for reflection over the line $y = x$

1 mark

Question 41

a) Determine the remainder when $3x^3 + 5x^2 - 13x - 3$ is divided by $(x + 3)$.

b) Is $(x + 3)$ a factor of $3x^3 + 5x^2 - 13x - 3$? Explain your reasoning.

Solution

a) $3(-3)^3 + 5(-3)^2 - 13(-3) - 3$

$$3(-27) + 5(9) - 13(-3) - 3$$

$$0$$

or

-3	3	5	-13	-3
		-9	12	3
	3	-4	-1	0

The remainder is 0 .

1 mark for remainder theorem

1 mark for remainder theorem

1 mark for synthetic division

(or any equivalent strategy)

1 mark

1 mark for explanation

1 mark

Question 42

State the range, the y -intercept, and the equation of the asymptote of the exponential function,

$$f(x) = 3^{x-1} + 2.$$

Range:

y -intercept:

Equation of the asymptote:

Solution

Range: $(2, \infty)$

1 mark for range

y -intercept: $\frac{7}{3}$

1 mark for y -intercept

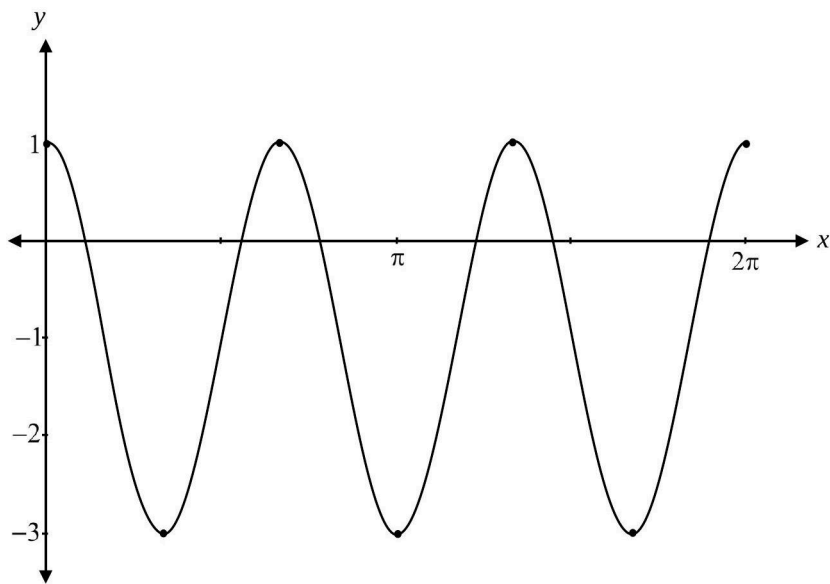
Equation of the asymptote: $y = 2$

1 mark for equation of the asymptote

3 marks

Question 43

The graph of $y = 2 \cos(3x) - 1$ below can be used to solve the equation $0 = 2 \cos(3x) - 1$ over the interval $[0, 2\pi]$. Indicate on the graph where to find at least one solution to the equation $0 = 2 \cos(3x) - 1$.



Solution

See arrows on graph above. 1 mark for indicating a solution on graph

1 mark

Note:

- Students do not need to show all six arrows.